



Setting the stage for developing deep learning tools
that support age estimation using images of otoliths
and other hard parts

DFO Dots and DFO Bot

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TESA - New Approaches to Fish Aging for Stock Assessments

May 13 2026

2.5 mm

Outline

1. Prologue
2. DFO Dots
3. DFO Bot
4. Some ideas for discussion
5. Epilogue

Prologue

Use of age estimates in stock assessment

- length observations are easier to obtain than age estimates
- age-based assessment models require catch-at-age matrices to perform demographic accounting and estimation of parameters
- catch-at-age matrices must be unbiased representations of the removals and of the age-structure of the population under scrutiny
- propagation of error and inclusion of uncertainty in age estimates

Best practices

TESA 2023 "best practices"
workshop

proceedings published in 2024

Proceedings of the Technical Expertise in Stock Assessment (TESA) national workshop on 'Best practices in age estimation', 31 January to 02 February 2023 in Moncton, New Brunswick

Daniel Ricard, Peter Comeau, Aaron Adamack, Jacob Burbank, Abby Daigle, Allan Debertin, Kim Emond, Tracey Loewen, Andrea Perreault, Gregory Puncher, Karen Robertson, Nicolas Rolland, Meredith Schofield, Andrew Smith, François-Étienne Sylvain and Stephen Wischniowski

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2024

**Canadian Technical Report of
Fisheries and Aquatic Sciences 3609**

Best practices

Draft report of Standard Operating Procedures for Gulf Region

- [inventory](#)
- otolith preparation
- reference collections
- calibration using reference collections
- verification of age estimates
- taking good and consistent images of otoliths
- age validation using bomb radiocarbon assays

Standard Operating Procedures for the estimation of marine fish ages in the Gulf Region of Fisheries and Oceans Canada

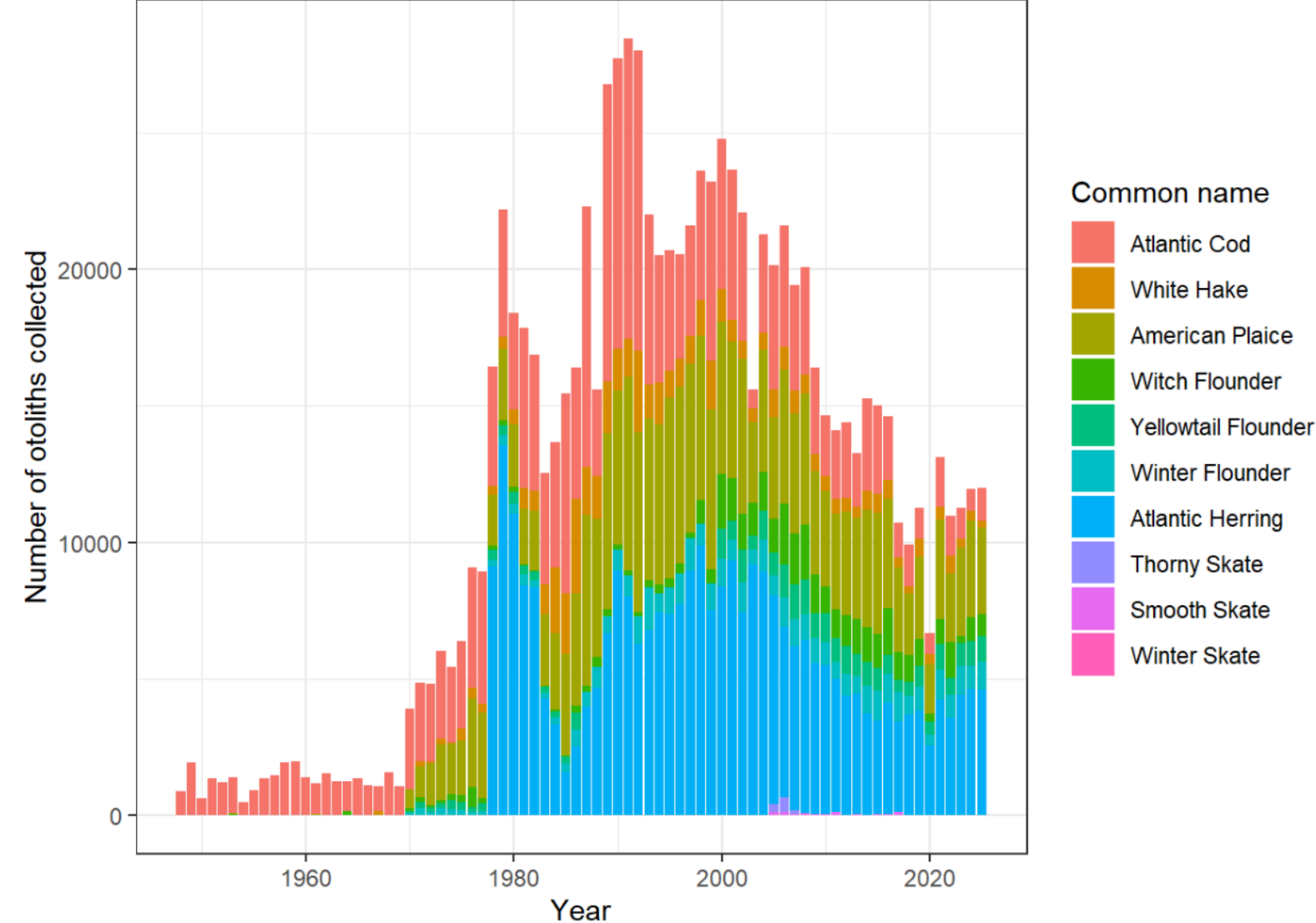
Daniel Ricard, Eliane Aubry, Cassidy Bertin, Jacob Burbank, Ian Copps, Andrew Darcy, Isabelle Forest, Laurie Maynard, Karen Robertson, Marie-France Robichaud, Sylvie Robichaud, Joeleen Savoie, and Carly Weber

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2026

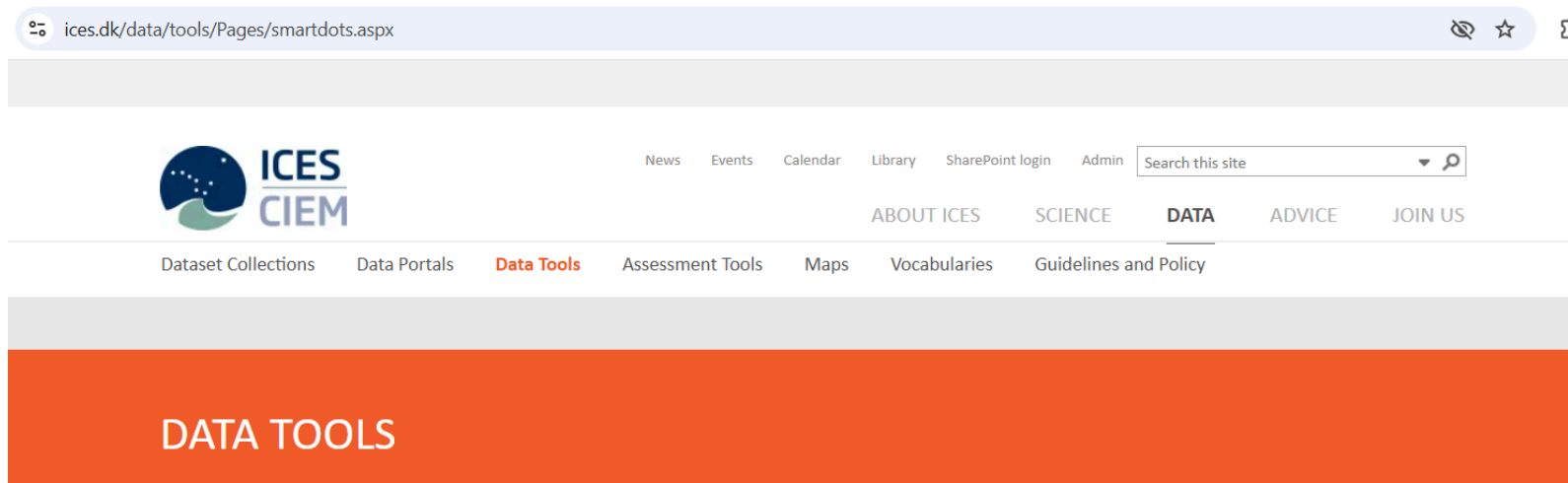
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Inventory of otoliths in the Gulf Region



SmartDots

SmartDots platform facilitates exchanges, workshops and training events for biological readings based on images



DATA BASICS

› Submission status

› Data calls

› Data formats

› Code lookup

› Code requests

QUALITY CONTROL

› Data Validation

SmartDots

SmartDots platform facilitates exchanges, workshops and training events for biological readings based on images

A set of software tools supports the user in managing all data of ICES biological reading (like age, maturity, larva identification) workshops and exchanges. The workshop or exchange manager can manage the meta data related to workshops and exchanges, and the biological readers can carry out readings by annotating images provided by the workshop managers. All registered data are available in the connected reporting environment.

Print it Send to f t in Share it

smartdots

VIEW EVENTS

MANAGE EVENTS (LOGIN)

DFO Dots

DFO Dots

an application to support age estimation using hard parts, hosted on DFO's dmapps, support the SmartDots API

basic workflow:

- create collection
- add specimens to collection
- add images to a folder (often on a NAS), and link images to specimens
- annotate images

other workflows:

- collection where age estimates and images are already available
- collection where age estimates are already available, and images are not necessarily available
- reference collection (training and calibration)
- validation study using bomb radiocarbon assays

DFO Dots within dmapps

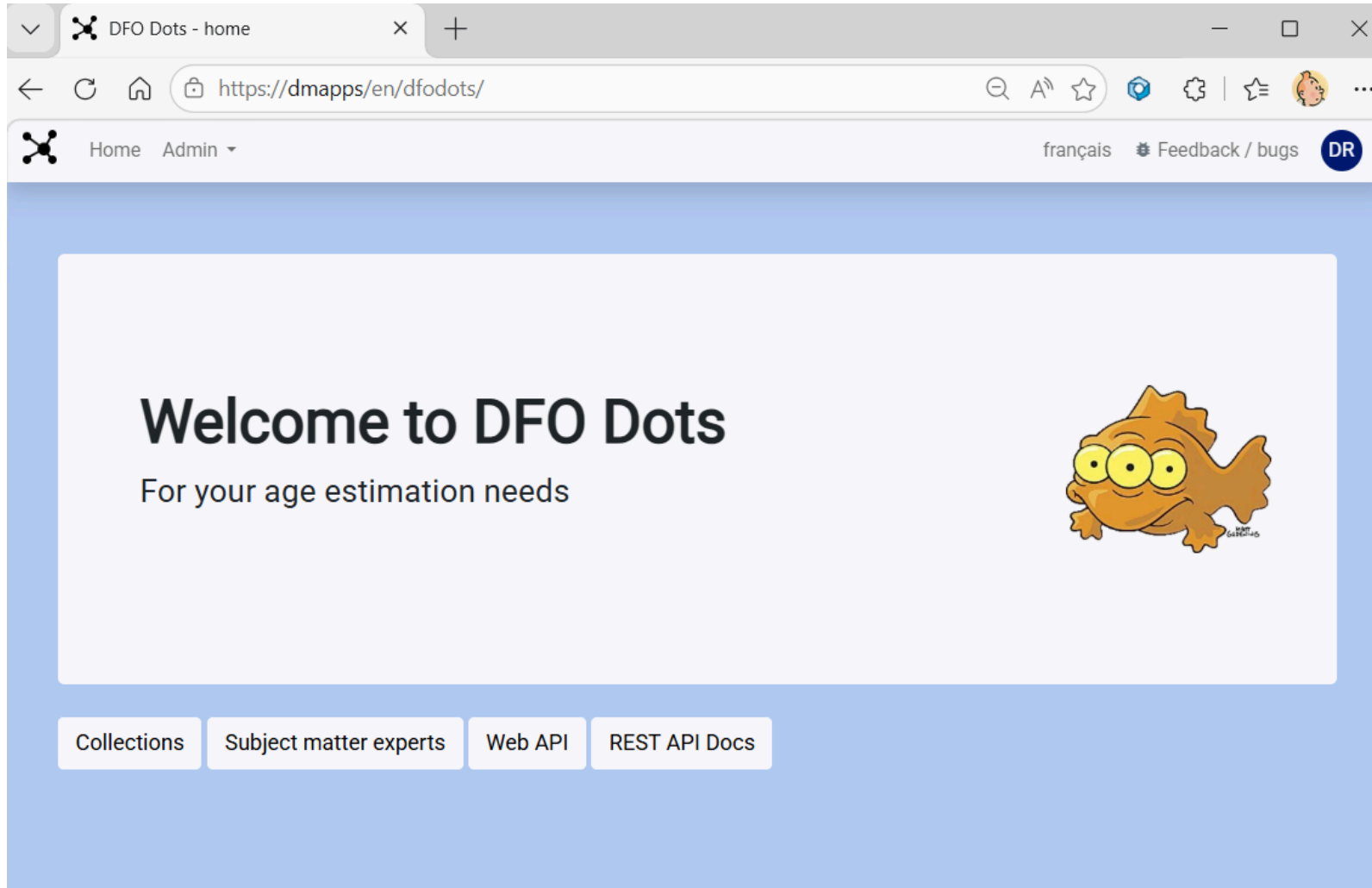
The screenshot shows the DM Apps web portal. The browser address bar displays `https://dmapps/en/`. The page header includes the DM Apps logo, a language selector set to 'français', a 'Feedback / bugs' link, and a user profile icon labeled 'DR'.

On the left sidebar, there is a 'Search apps' input field. Below it, a 'Contact Us' section dated 'Jan. 25, 2022, 12:01 p.m.' provides a generic inbox email: DFO.DMApps-ApplisGD.MPO@dfo-mpo.gc.ca. A note states: 'If you have any general questions or comments, please do not hesitate to reach out to the team.' Below this, an 'Unclassified Information' section dated 'Jan. 1, 2020, 10:28 a.m.' includes a disclaimer: 'Please note that this site is only intended for the storage of **unclassified information only**.' At the bottom of the sidebar, it says 'Your last login was recorded on April 14, 2026 at 1:52 p.m.'

The main content area displays a grid of application tiles:

- Cruises**: Collection of regional cruise metadata and oceanographic data.
- CARL**: Catch Register for the Atlantic.
- Science Data Inventory**: Tool for organizing and managing regional data resources.
- AAHPU** (AAH Processing Unit): AAH Processing Unit.
- Biological Sampling Information System (BSIS)**: Data archive for biological sampling data.
- Gulf Science Field Planning Tool**: Tool used for the planning and coordination of field work.
- CAT App**: App for CATs.
- Canadian Science Advisory Secretariat**: Tool for tracking meetings, requests and publications.
- DFO Dots**: Collection of ageing structures and WebAPI for ICES SmartDots. This tile is circled in red.
- DFO stocks**: App to handle DFO stocks and assessments.
- Marine Diets**: Stomach contents analysis database / application.

DFO Dots splash page



DFO Dots

costs

- creating a collection using a CSV file of suitable format
- taking (and storing) good and consistent images

benefits

- CRUD (create, read, update, delete)
- discoverability and archival
- application is developed in-house, features can be implemented and improved
- available nationally from dmapps
- compatible with SmartDots, elegantly transferable from/to ICES infrastructure

DFO Dots American Plaice collections

DFO Dots - Collections

https://dmapps/en/dfodots/collections/?id=&institute__region=&cat...

Home / Collections

Collections

Search (name, description)

Institute region

Species

Project lead(s)

Material type

Filter

Clear

(click on a header to sort current page)

New

Region	Name	Description	Project lead(s)	Catch	Material type	Specimens (annotations)	Percent complete
Gulf	American Plaice NAFO Div. 4T - RV 2024	American Plaice otoliths collected during the 2024 September survey in the southern Gulf of St. Lawrence	Daniel Ricard, Joeleen Savoie	Hippoglossoides platessoides	Otolith	3516 (0)	0%
Gulf	American Plaice NAFO Div. 4T - RV 2022	American Plaice otoliths collected during the 2022 September survey of the southern Gulf of St. Lawrence. Specimens were collected on both CCGS Teleost and CCGS Capt. Jacques Cartier.	Daniel Ricard, Karen Robertson	Hippoglossoides platessoides	Otolith	2523 (128)	99%
Gulf	American Plaice gonads collected for histology	This collection contains the American Plaice specimens from which gonads were photographed and collected for histological analyses.	Daniel Ricard, Kadra Benhalima, Joeleen Savoie	Hippoglossoides platessoides	Gonad	24 (1)	0%
Gulf	American Plaice NAFO Div. 4T - RV 2025	American Plaice otoliths collected during the 2025 September survey in the southern Gulf of St. Lawrence	Daniel Ricard	Hippoglossoides platessoides	Otolith	3182 (0)	0%
Gulf	American Plaice NAFO Div. 4T - RV 2021	American Plaice otoliths collected during the 2021 September survey in the southern Gulf of St. Lawrence	Daniel Ricard, Joeleen Savoie	Hippoglossoides platessoides	Otolith	3631 (122)	84%
Gulf	American Plaice NAFO Div. 4T - RV 2023	American Plaice otoliths collected during the 2023 September survey in the southern Gulf of St. Lawrence	Daniel Ricard	Hippoglossoides platessoides	Otolith	3252 (102)	3%
Gulf	American Plaice NAFO Div. 4T reference collection	Otolith reference collection for southern Gulf of St. Lawrence American Plaice (NAFO Div. 4T).	Daniel Ricard	Hippoglossoides platessoides	Otolith	255 (757)	100%

displaying 7 of 7

DFO Dots American Plaice 2021 otoliths

DFO Dots - American Plaice NAFO

https://dmapps/en/dfodots/collections/46/view/

Home

Admin

français

Feedback / bugs

DR

[Home](#)

[Collections](#)

American Plaice NAFO Div. 4T - RV 2021

American Plaice NAFO Div. 4T - RV 2021

EditCloneDelete

PlotsReports

Region	Gulf
Institute	Gulf Fisheries Centre (Gulf)
Description	American Plaice otoliths collected during the 2021 September survey in the southern Gulf of St. Lawrence
Project lead(s)	Daniel Ricard, Joeleen Savoie
Annotators	Karen Robertson, Joeleen Savoie
Species	American plaice (Hippoglossoides platessoides)
Material type	Otolith
ICES preparation method	Whole in glycerin (ALG)
Folder path	https://glfscidm002/otoliths/American-plaice/2021-rv-imaged-otoliths
Is reference collection?	No
Specimen progress (completed / total)	3063 / 3631 (84%)
Annotation progress (completed / total)	0 / 0 (0%)

Specimen Map

DFO Dots American Plaice 2021

sagittal otoliths from a 20 year old female

Home Admin

français Feedback / bugs DR

Home / Collections / Collection / Hippoglossoides platessoides TEL-221-116-2249

Hippoglossoides platessoides TEL-221-116-2249

COMPLETE

Edit Delete

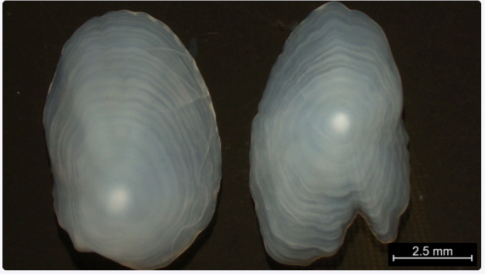
← Previous Specimen (Page Up)

Next Specimen (Page Down) →

Next Incomplete Specimen (Shift + Page Down) →

Collection date	2021-09-17
Sample origin	Sea
Survey identifier	TEL-2021-221
Sample identifier	Set 116
Specimen identifier	TEL-221-116-2249
Specimen identifier type	fish unique ID
Area identifier	NAFO 4T
Length (mm)	430.0
Length type	Fork
Weight (g)	648.0
Sex	Female
Fish number	2249
Dot count(s)	

Image 1 - TEL-221-116-2249-photo-1.jpg (0 annotations)



This image has no annotations

New Annotation +

Image Detail

Image 2 - TEL-221-116-2249-photo-2.jpg (0 annotations)



DFO Dots American Plaice reference collection

comparing annotations from a 21 year old female

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Comparing Annotations of Image 1

+

-

Isabelle Forest (2023-04-06)

Joeleen Savoie (2023-04-06)

Karen Robertson (2023-04-06)

Annotations of Image 1

Ager	Code	Edge	Quality	Dot count	Authoritative	Complete
Isabelle Forest	OWR	Opaque	AQ1	21	✓	✓
Joeleen Savoie	OWR	Opaque	AQ2	20	✗	✓
Karen Robertson	OWR	Opaque	AQ2	21	✗	✓

Image details:

Uuid: 981e3eb6-30de-4809-afe6-c55ce3540261

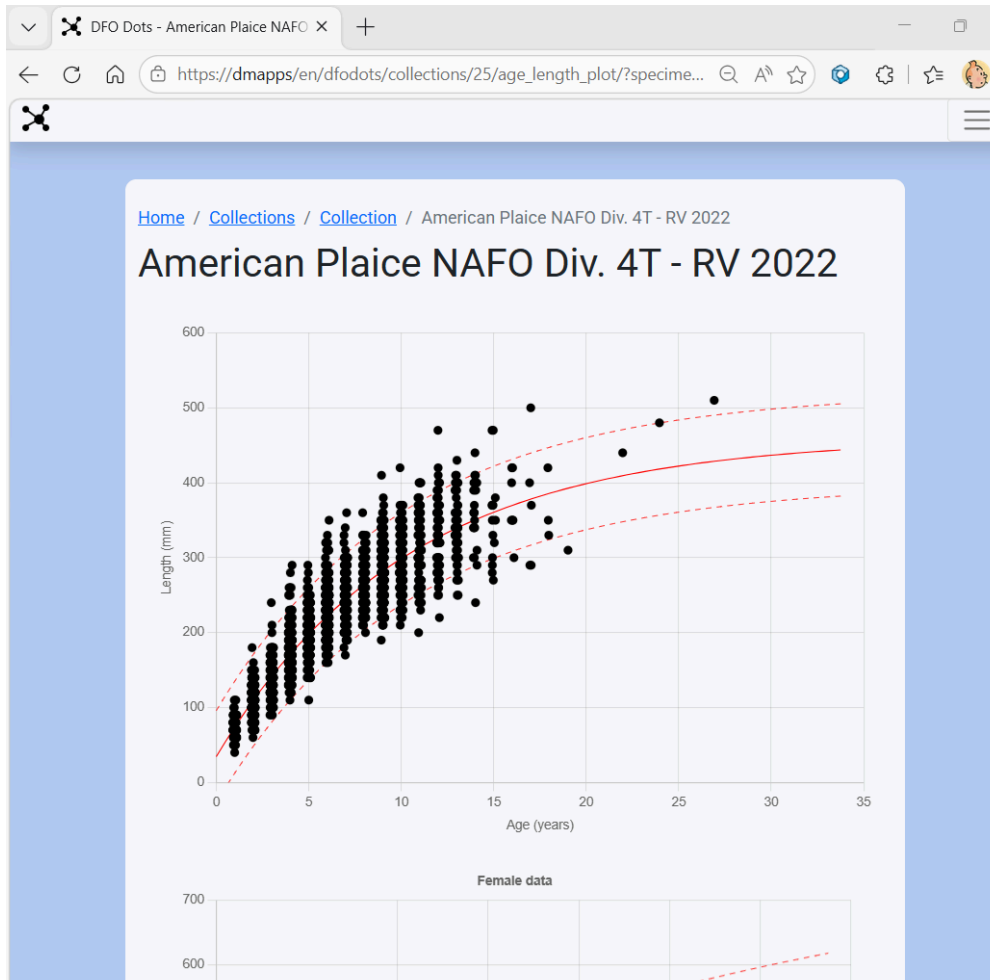
Image scale (px/mm): 150

Filename: 1994-COMM-40-0096-photo-0001-image-scale-150.JPG

[\(Edit Image Metadata\)](#)

DFO Dots American Plaice 2022

length-weight plot used for verification



DFO Bot

DFO Bot

a deep-learning tool trained to analyse and classify otolith images (following Moen et al. 2018, Politikos et al. 2021 and Sigurðardóttir et al. 2023)

- developed in Python using the ResNet50 package, development tree on GitHub
- training using otolith images, initially American Plaice (*Hippoglossoides platessoides*) and Atlantic Herring (*Clupea harengus*)
- uses otolith images and age estimates that are on DFO Dots
- predictions of ages using images

- Côme yesterday: time to move on from "image to age estimate", and start using annotations, probabilistic approach to predictions
- DFO Bot is on its second development cycle and is moving in a similar direction

ideas for discussion

Topics for discussion

- neural networks for single species or multiple species?
- moving from the "black box", how to get feedback from the neural network (saliency mapping)?
- frequency of model updates, criteria and thresholds for performance
- novel methods are not replacing skilled personnel, they are supporting them: how to best use these novel tools?
- DFO High Power Computing: GPU available for CNN training
- demand-side management: how many otoliths does one need? how many age estimates?
- equipment, laboratory and analytical setups: how expensive and accessible is the required equipment? how difficult is it to conduct analyses? how to design good lab workspaces?

Epilogue

Where to from here?

- use of age estimates in stock assessment: how many otoliths do I need?
how to achieve an unbiased estimation of age structure?
- how to go from lengths to ages: workshop on catch-at-age calculations
that are transparent and replicable
- examining value of information from age estimation tasks: age-length
pairs versus annotations

Where to from here?

- use of DFO Dots for other hard parts (scales, mollusk shells, teeth, vertebrae, spines)
- use of DFO Dots for fish eggs and larvae, and for gonads and maturity staging
- accessing age estimates and annotations from DFO Dots for analyses (server-side reporting and API access)
- DFO Bot refinements (more images and age estimates available for Yellowtail Flounder (*Myxopsetta ferruginea*) and Winter Flounder (*Pseudopleuronectes americanus*))
- collaboration with ICES through [Smart4SAM](#)

Yet another draft report

Design goals, technical implementation and practical use of DFO Dots, an application to support age estimations using hard structures

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David Fishman and Jenni McDermid - CADI Gulf Region

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The image shows two scallop shells, likely from the genus Arctica, resting on a dark surface. The shells are light-colored with prominent concentric growth lines. A scale bar in the bottom right corner indicates a length of 2.5 mm.

2.5 mm

Thank you